METHODOLOGY PASSPORT CARD

Research Methodology & Experimental Design

1. PROJECT INFORMATION

- **Research Title:** Classification of Textual AI-Phishing Methods in Personal Communications of Private Users
- **Authors:** Tursynbayev Yerassyl, Ospankul Zharkyn
- **Repository URL:** https://github.com/yeraaaaaah/ai-phishing-classification-benchmark

2. RESEARCH TYPE & JUSTIFICATION

- **Research Design:** Mixed-Methods Research
- **Methodological Justification:**

  - **Quantitative Phase:** Structural-linguistic analysis of text properties, including error rate calculations per 100 words, readability metrics, and evaluation of formal vs. emotional tone proportions.
  - **Empirical Phase:** Human participant evaluation measuring spontaneous vigilance levels, trust scores on a 1–7 Likert scale, and action execution probabilities.

3. FORMULATION OF HYPOTHESES

Hypothesis Set 1: Linguistic & Stylistic Characteristics

- **Null Hypothesis (H_0):** Differences in linguistic coherence, syntactical error rates, and stylistic tone ratios between LLM-generated phishing texts and traditional manual phishing texts are statistically non-significant.
- **Alternative Hypothesis (H_1):** AI-generated phishing messages exhibit statistically higher linguistic coherence, fewer syntactical errors, and a higher proportion of formal domain-specific style compared to traditional manual phishing.

Hypothesis Set 2: Psychological Impact & User Vigilance

- **Null Hypothesis (H_0):** Text generation method and the presence or absence of linguistic anomalies ("red flags") have no statistically significant impact on user spontaneous vigilance, perceived authority, or action execution rates.
- **Alternative Hypothesis (H_1):** Users' spontaneous vigilance when evaluating AI-generated phishing messages is statistically lower, while the absence of linguistic anomalies strengthens the sender authority trigger, increasing action execution rates.

4. VARIABLE MATRIX

Variable Category	Variable Name	Description & Parameters
Independent Variables (IV)	**1. Generation Method** **2. Presence of Red Flags**	• LLM-generated (ChatGPT, Claude, Gemini) vs. Traditional Manual Writing • Presence vs. Complete absence of linguistic and stylistic anomalies
Dependent Variables (DV)	**1. Text Quality Metrics** **2. Stylistic Pattern Ratio** **3. Spontaneous Vigilance** **4. Action Execution Rate**	• Error count per 100 words, Readability Index • Proportion of formal business style vs. emotional threat markers • Detection accuracy (%) and trust rating (1–7 Likert scale) • Percentage of users attempting link clicks or credential entry
Controlled Variables (CV)	**Experimental Constants**	• Message length and topic domain • Target language (English) • Communication platform / channel • Participant digital literacy level and age group

5. METRICS & BASELINE SPECIFICATION

- **Primary Metrics:**
  - **Linguistic Metric:** Error Rate per 100 words & Flesch-Kincaid Readability Index.
  - **Empirical Metric:** Detection Accuracy Percentage (%).
- **Guardrail Metrics:**
  - **Execution Latency:** Benchmark execution time < 2000 ms per text sample.
  - **False Positive Rate:** Classification error rate for legitimate messages < 5%.
  - **Stability:** 0% unhandled exceptions during pipeline execution.
- **Baseline & Control Benchmark:**
  - **Control Corpus:** A curated set of traditional human-crafted phishing emails and messages gathered from open-source threat intelligence (OSINT) databases.
  - **Standard Detection Baseline:** Traditional rule-based and keyword-matching detection mechanisms.